

Microbiome Bioinformatics Portfolio

Reproducible Workflows for Soil and Plant-Associated Microbial Community Analysis

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1 Overview

This portfolio presents reproducible microbiome bioinformatics workflows for soil and plant-associated microbial communities using amplicon sequencing data. The analytical framework integrates preprocessing, ASV filtering, alpha diversity, beta diversity, relative abundance profiling, and publication-ready visualization.

These workflows were developed to support microbial community studies based on 16S rRNA and ITS amplicon datasets and to provide reproducible outputs for ecological interpretation, scientific reporting, and consulting-oriented deliverables.

2 Professional Scope

These workflows demonstrate technical expertise in:

- Amplicon sequencing data analysis
- QIIME2-derived data integration in R
- ASV filtering and preprocessing
- Alpha diversity analysis
- Beta diversity analysis
- PERMANOVA and dispersion testing
- Relative abundance profiling
- Publication-ready figure generation

3 Computational Environment

All analyses were performed in R using QIIME2-derived datasets on macOS.

Core tools used in these workflows include:

- QIIME2
- qiime2R
- phyloseq
- vegan
- tidyverse
- microbiome
- patchwork
- openxlsx
- rstatix
- ggplot2
- Inkscape

4 Workflow Applications

The portfolio includes two main application areas:

4.1 Soil microbiome analysis

This workflow supports bacterial and fungal community analysis from soil-associated microbiome datasets using 16S rRNA and ITS amplicon sequencing data.

4.2 Plant-associated microbiome analysis

This workflow supports the analysis of microbial communities associated with plant tissues, including endophytic and plant-associated bacterial datasets, using comparable preprocessing, diversity, and taxonomic profiling strategies.

5 Shared Workflow Structure

Both soil and plant-associated workflows follow the same general analytical structure:

1. Import of QIIME2-derived feature tables, taxonomy, trees, and metadata
2. Construction of phyloseq objects
3. Sample filtering by sequencing depth
4. Taxonomic filtering and cleaning
5. Generation of rarefied and non-rarefied objects
6. Alpha diversity analysis
7. Beta diversity analysis
8. Relative abundance profiling
9. Export of figures and statistical summary tables

6 Use Case 1: Soil Microbiome Workflow

The soil microbiome workflow was designed for microbial community analysis across two study regions, Caldas and Risaralda, and includes both bacterial (16S rRNA) and fungal (ITS) datasets.

6.1 Main analytical components

- ASV filtering and preprocessing
- Alpha diversity analysis using Observed richness, Shannon, and Chao1
- Beta diversity analysis using Bray-Curtis, Jaccard, and UniFrac for 16S
- PERMANOVA and betadisper
- Relative abundance profiling using stacked bar plots
- Targeted taxonomic comparisons using Kruskal-Wallis and Dunn tests

6.2 Example Output:

6.2.1 Alpha Diversity

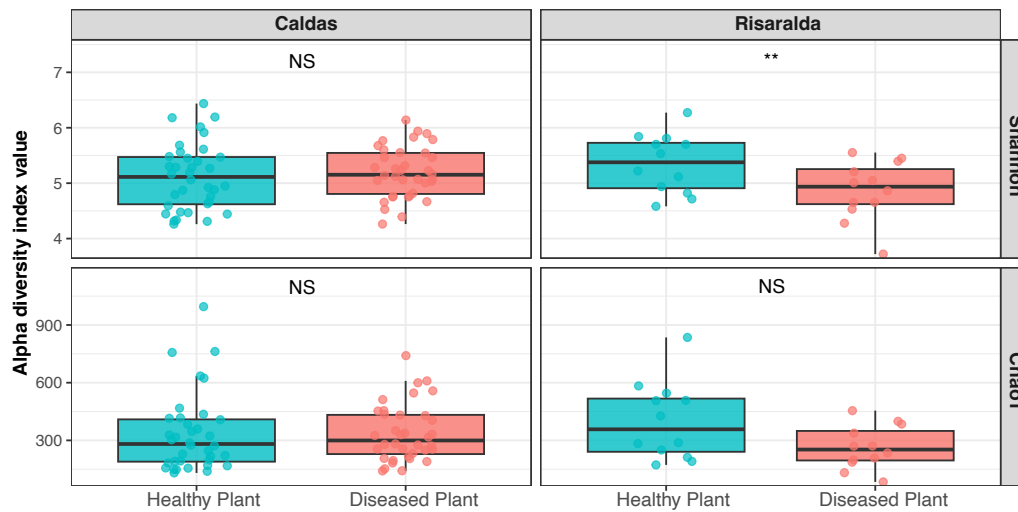


Figure 1: Alpha diversity comparison across plant conditions.

6.2.2 Beta Diversity Ordination

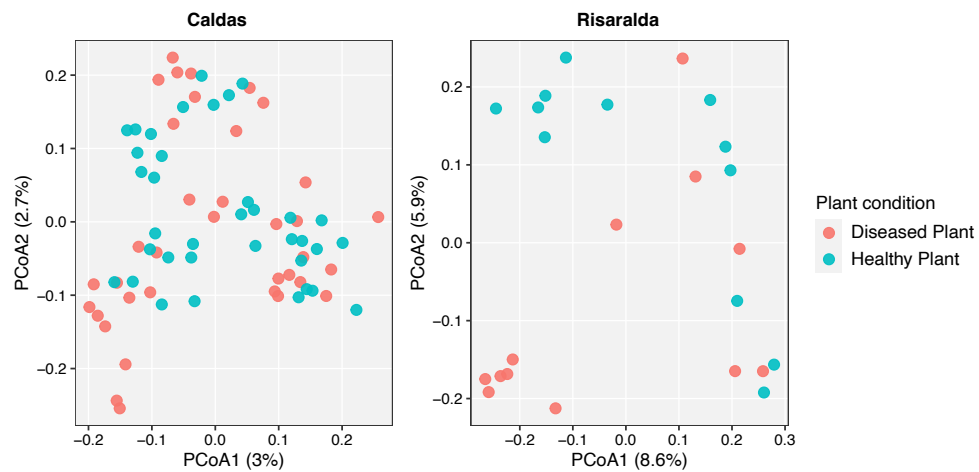


Figure 2: PCoA ordination based on Bray-Curtis distance.

6.2.3 Relative Abundance

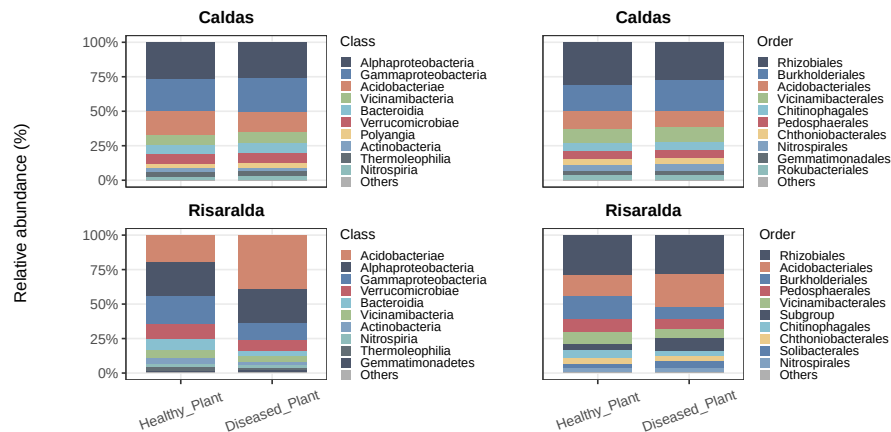


Figure 3: Taxonomic composition of microbial communities across samples.

6.3 Main deliverables

- Filtered phyloseq objects
- Rarefied phyloseq objects
- Alpha diversity plots
- Beta diversity ordination plots
- PERMANOVA summary tables
- Relative abundance figures
- Publication-ready multipanel visualizations

7 Use Case 2: Plant-Associated Microbiome Workflow

The plant-associated workflow applies the same analytical logic to microbial communities associated with plant tissues. This includes preprocessing, diversity analysis, taxonomic profiling, and statistical comparisons across biological conditions or sample groups.

7.1 Main analytical components

- ASV filtering and preprocessing
- Metadata harmonization
- Alpha diversity analysis
- Beta diversity analysis
- Relative abundance profiling
- Differential comparison of microbial groups
- Generation of figures for scientific manuscripts

7.2 Main deliverables

- Standardized phyloseq objects
- Diversity summary plots
- Ordination analyses
- Relative abundance plots
- Multipanel scientific figures
- Statistical summary tables

8 Typical Deliverables

These workflows generate outputs suitable for research projects, manuscripts, and consulting reports, including:

- Filtered and rarefied microbiome objects
- Diversity plots
- Ordination figures
- PERMANOVA and dispersion tables
- Taxonomic composition summaries
- Publication-ready figures
- Excel summary tables

9 Professional Relevance

These workflows are suitable for:

- Remote microbiome data analysis roles
- Freelance scientific consulting
- Collaborative microbial ecology projects
- Technical support for microbiome studies
- Publication-oriented data analysis and figure generation

This portfolio reflects practical expertise in microbiome bioinformatics, microbial ecology, soil microbiology, plant-associated microbial communities, and scientific data visualization.

10 Final Remarks

This portfolio presents a modular and reproducible microbiome bioinformatics framework applicable to both soil and plant-associated microbial community datasets. The workflows combine statistical rigor, biological interpretation, and reproducible reporting, making them suitable for scientific collaboration, consulting services, and remote bioinformatics work.